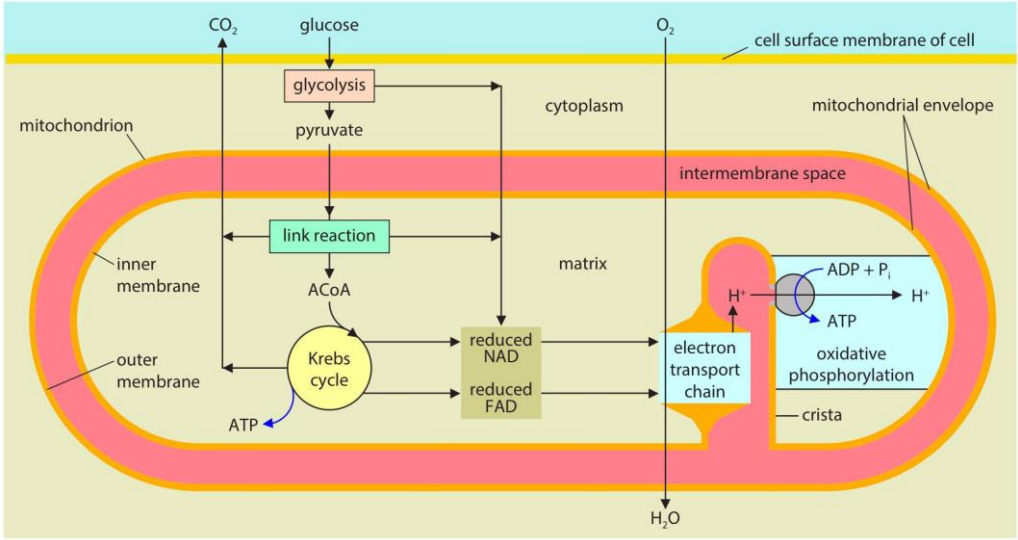


Unit 5: Respiration, Internal Environment, Coordination and Gene Technology

Topic 7 – Respiration, Muscles and the Internal Environment

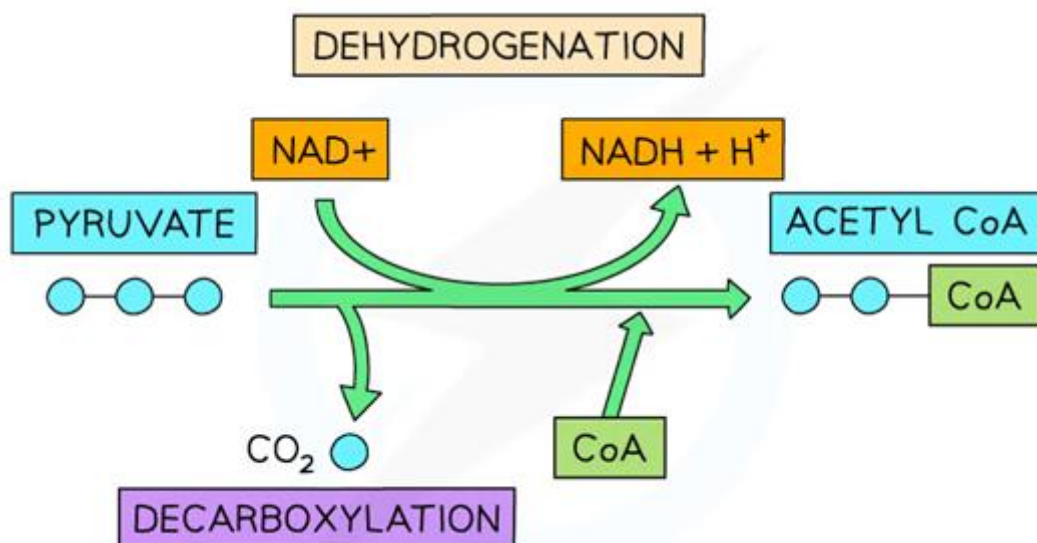
Candidates will be assessed on their ability to:

7.1	(i) understand the overall reaction of aerobic respiration as splitting of the respiratory substrate to release carbon dioxide as a waste product and reuniting hydrogen with atmospheric oxygen with the release of large amounts of energy (ii) understand that respiration is a stepped process, with each step controlled and catalyzed by a specific intracellular enzyme <i>Names of specific enzymes are not required.</i>
i	Overall Reaction and Process of Aerobic Respiration: glucose + oxygen → carbon dioxide + water + energy $C_6H_{12}O_6 + 6 O_2 \rightarrow 6 CO_2 + 6 H_2O + 2870kJ$  The diagram illustrates the process of aerobic respiration within a cell and a mitochondrion. Glucose enters the cell and undergoes glycolysis in the cytoplasm to produce pyruvate. Pyruvate then enters the mitochondrion and undergoes the link reaction in the matrix to produce Acetyl CoA (ACoA). ACoA enters the Krebs cycle, which releases CO₂ and produces reduced NAD and reduced FAD. These reduced co-factors enter the electron transport chain on the inner mitochondrial membrane. Oxygen (O₂) enters the cell and the mitochondrion, where it is reduced to water (H₂O). The electron transport chain also drives oxidative phosphorylation, where ADP + P_i is converted to ATP. The diagram also shows the structure of the mitochondrion, including the outer membrane, inner membrane, intermembrane space, and crista.
ii	Factors that affect rate of respiration: - Respiration is enzyme-controlled reaction so factors that affect enzyme activity will also affect rate of respiration. Enzymes in Respiration: 1. Decarboxylase: catalyzes decarboxylation in linked reaction to remove carbon dioxide 2. Dehydrogenase: catalyzes dehydrogenation in linked reaction to remove hydrogen Coenzymes in Respiration: - Coenzymes are small molecules that assists in enzyme-catalyzed reaction 1. NAD (Nicotinamide Adenine Dinucleotide): hydrogen carrier

	2. FAD (Flavin Adenine Dinucleotide): hydrogen carrier Process of Respiration: 1. Glycolysis: Happens in cytoplasm 2. The Link Reaction: Happens in matrix of mitochondria 3. The Krebs Cycle: Happens in matrix of mitochondria 4. Oxidative Phosphorylation: Happens at the inner membrane of mitochondria
7.2	understand the roles of glycolysis in aerobic and anaerobic respiration, including the phosphorylation of hexoses, the production of ATP by substrate level phosphorylation, reduced coenzyme, pyruvate and lactate <i>Details of intermediate stages and compounds are not required.</i>
	KEY ● = CARBON ● = PHOSPHATE The diagram illustrates the process of glycolysis. It starts with a box labeled 'GLUCOSE' containing six blue circles representing carbon atoms, labeled '(6C)'. A downward arrow leads to a box labeled '2 x TRIOSE PHOSPHATE' containing three blue circles and one orange circle representing a phosphate group, labeled '(3C)'. This step is labeled 'PHOSPHORYLATION' and shows '2 ATP' being used to become '2 ADP'. From the triose phosphate box, another downward arrow leads to a box labeled '2 x PYRUVATE' containing three blue circles, labeled '(3C)'. This step is labeled 'DEPHOSPHORYLATION' and shows '4 ADP' being converted to '4 ATP' (labeled '4 ATP PRODUCED'). A side arrow from the triose phosphate box to the pyruvate box is labeled 'OXIDATION' and shows '2 NAD' being converted to '2 NADH⁺'. Copyright © Save My Exams. All Rights Reserved Glycolysis: 1. Phosphorylation: - Glucose is converted into Glucose-6-Phosphate by using energy from ATP - Glucose-6-Phosphate is converted into (2) Triose Phosphate molecules by using energy from ATP 2. Oxidation: - Each Triose Phosphate is then oxidized to Glycerate-3-Phosphate - This is carried out by removing Hydrogen atoms which is accepted by NAD to form reduced NAD 3. Dephosphorylation: - The Phosphate group is transferred to ADP to produce ATP and also Pyruvate Products of Glycolysis: → For one glucose molecule / one run of glycolysis: 1. (2) Pyruvate 2. Net (2) ATP molecules 3. (2) Reduced NAD

7.3

understand the role of the link reaction and the Krebs cycle in the complete oxidation of glucose and formation of carbon dioxide (CO₂) by decarboxylation, ATP by substrate level phosphorylation, reduced NAD and reduced FAD by dehydrogenation (names of other compounds are not required) and that these steps take place in mitochondria, unlike glycolysis which occurs in the cytoplasm



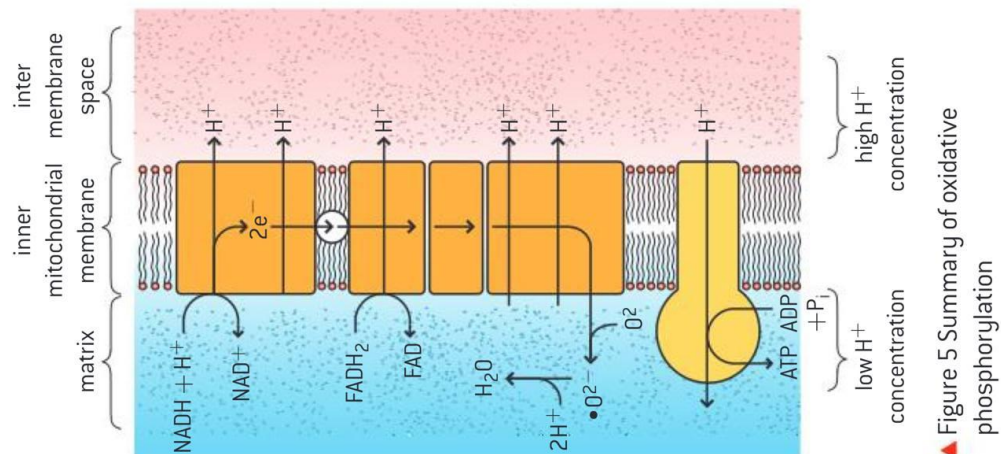
The Link Reaction:

1. **Decarboxylation:** Pyruvate is decarboxylated and one molecule of carbon dioxide is produced.
2. **Dehydrogenation:** The two-carbon intermediate is oxidized. Hydrogen atoms are removed and transferred to the coenzyme NAD to reduced NAD.
3. The resulting acetyl group combines with Coenzyme A (CoA) to form **Acetyl CoA**, which then enters the Krebs cycle.



Products of The Link Reaction:

- For one run of Link reaction:
 1. (1) Acetyl CoA
 2. (1) Carbon Dioxide
 3. (1) Reduced NAD
- For one glucose molecule:
 1. (2) Acetyl CoA
 2. (2) Carbon Dioxide
 3. (2) Reduced NAD



Oxidative Phosphorylation

Role of Chemiosmosis in synthesis of ATP:

1. Hydrogen atoms are transported to the electron transport chain
2. This is done by coenzyme NAD and FAD
3. Electrons pass along the electron transport chain, releasing energy from a series of redox reactions
4. This is used to move protons to the intermembrane space
5. Protons diffuse back into the mitochondrial matrix through ATP synthase on stalked particles
6. ATP synthase catalyzes formation of ATP from ADP and Pi

Role of Oxygen:

- Oxygen is final electron acceptor
- Oxygen forms water by using electron from ETC and hydrogen ions that passes through ATP synthase
- This allows to maintain proton gradient across the inner mitochondrial membrane so chemiosmosis can continue

Overall: For one glucose molecule:

Stage	ATP used	ATP made	Net gain in ATP
glycolysis	2	4	2
link reaction	0	0	0
Krebs cycle	0	2	2
oxidative phosphorylation	0	28	28
Total	2	34	32

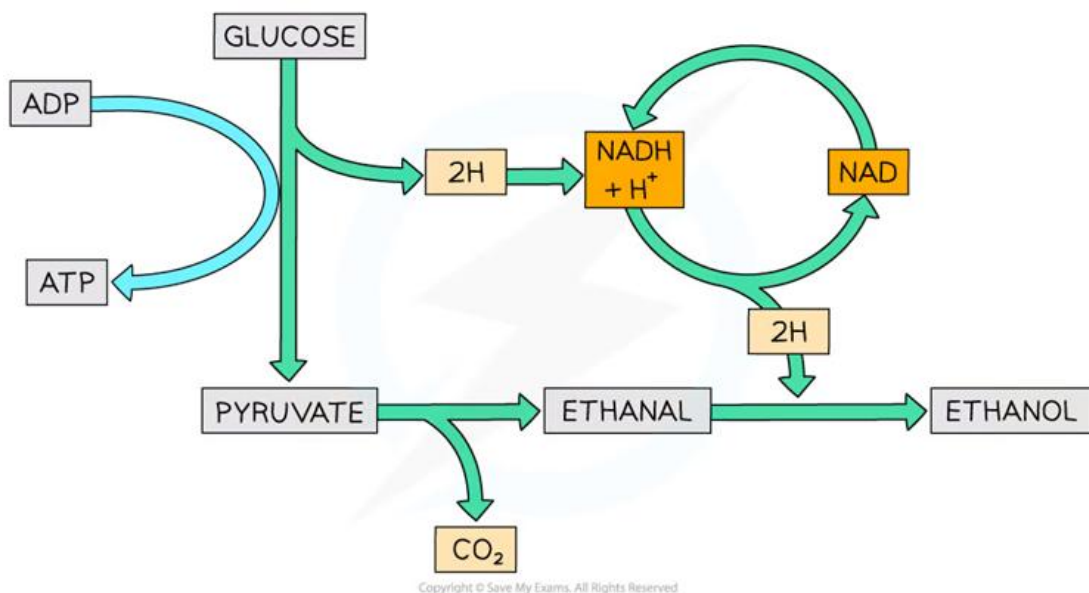
7.5 understand what happens to lactate after a period of anaerobic respiration in animals

Anaerobic Respiration:

- Without oxygen inside mitochondrion, there is nothing to accept the electrons at the end of the electron transport chain.
- So, no free carrier in the chain to accept hydrogen from reduced NAD and reduced FAD so they remain reduced.
- This stops the Krebs cycle as no oxidized NAD or FAD to enable dehydrogenation steps to take place.
- However, cells have evolved in (2) different ways to continue glycolysis and to produce ATP from substrate level phosphorylation

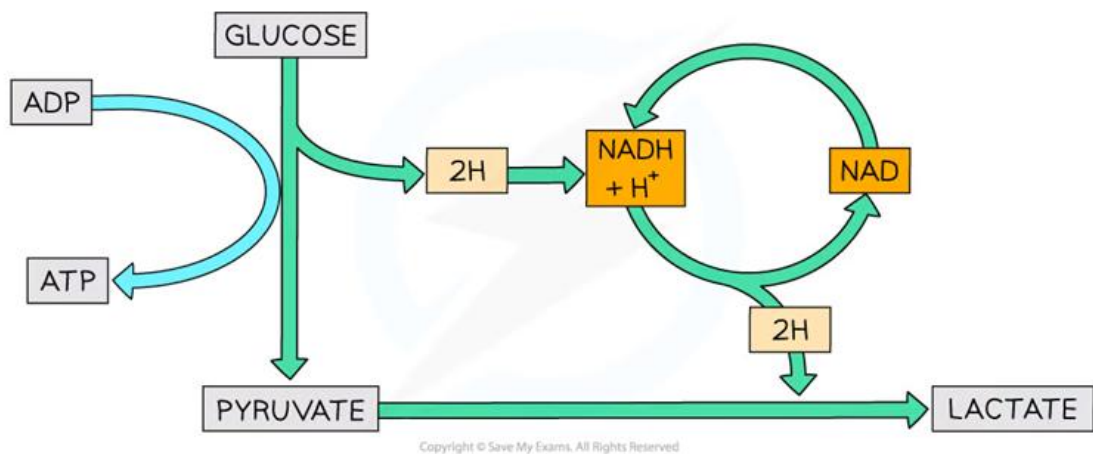
1. Ethanol Fermentation

2. Lactate Fermentation



1. Ethanol Fermentation

- Pyruvate is decarboxylated to ethanal; This produces one CO₂ molecule
- Ethanal is reduced to ethanol by enzyme alcohol dehydrogenase
- Also, hydrogen from reduced NAD is passed to ethanal
- Ethanol cannot be further metabolized and it is simply a waste product

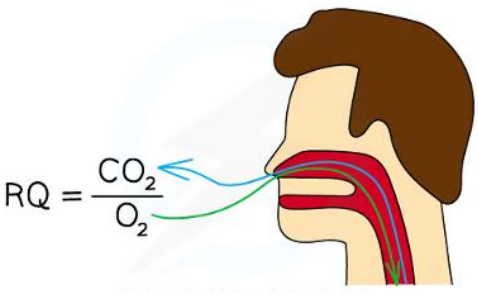
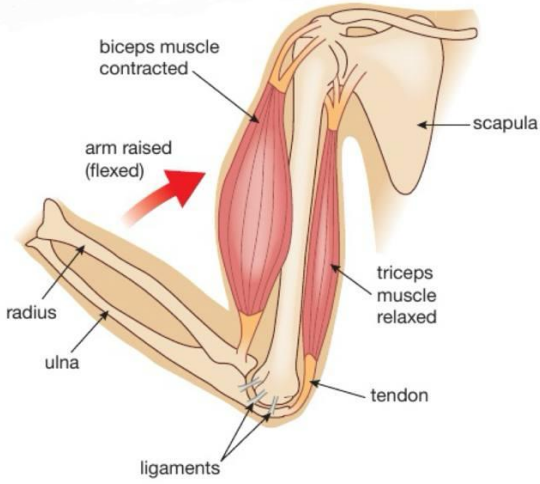
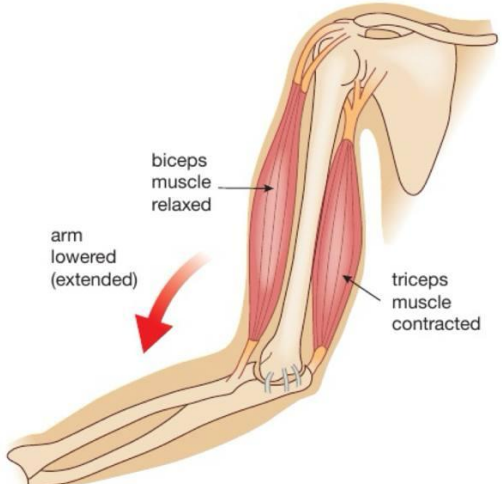


2. Lactate Fermentation

- Pyruvate acts as hydrogen acceptor and is converted to lactate by enzyme lactate dehydrogenase
- Lactate can be oxidized and convert back to pyruvate which can be fed again into the Krebs cycle to run oxidative phosphorylation
- → Alternatively, lactate can be converted to polysaccharides glycogen and stored
- → These processes happen in liver in mammals
- → The oxidation of lactate requires extra oxygen (oxygen debt), and this is why we continue to breathe more deeply and heavily than usual after exercise has finished

Glycogen → Glucose → Pyruvate Pathway:

- This process produces net (3) ATP molecules as it bypasses the phosphorylation and oxidation by just using (1) ATP and the dephosphorylation produces (4) ATP so (3) ATP net.

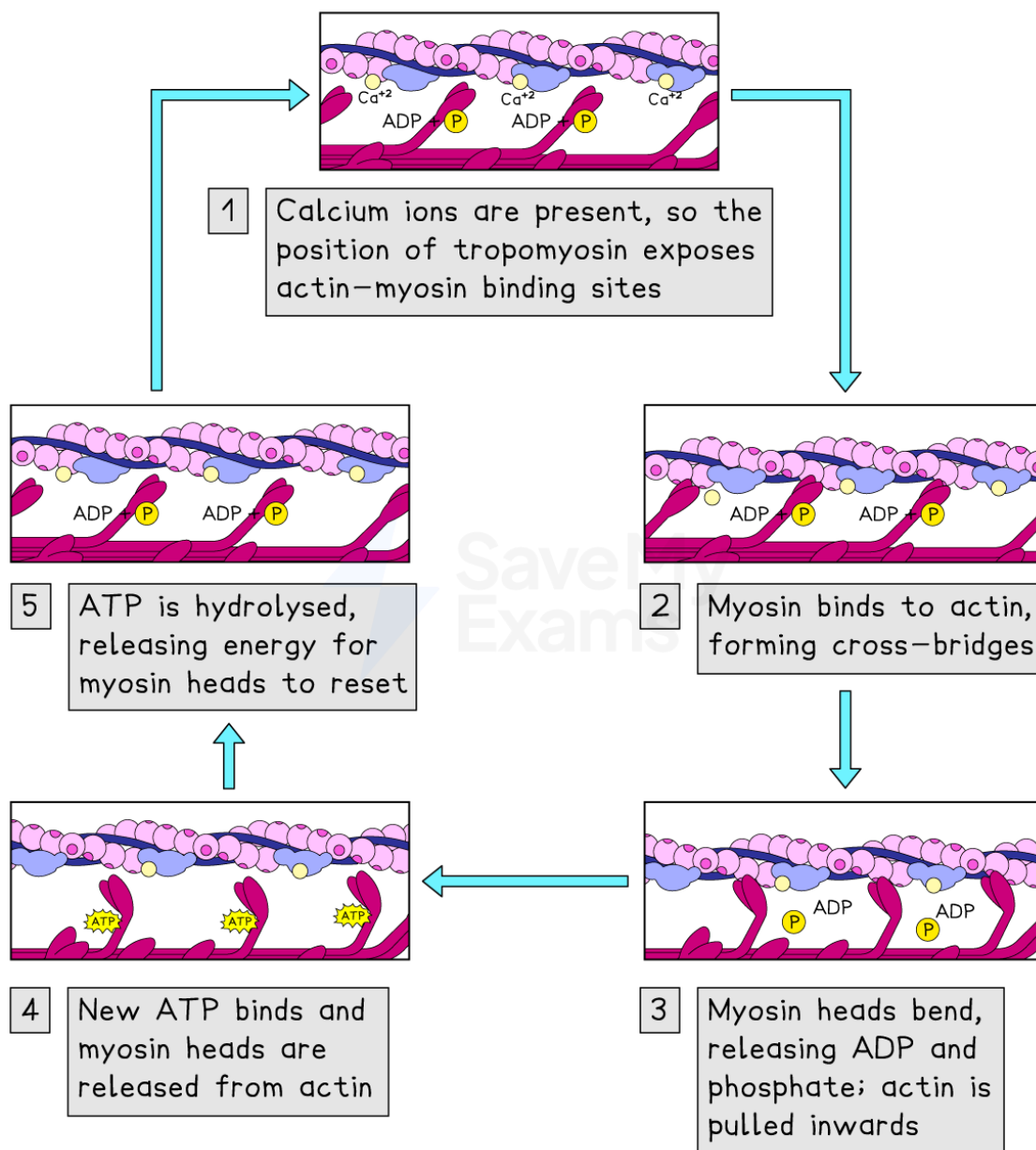
7.6	understand what is meant by the term <i>respiratory quotient (RQ)</i> $RQ = \frac{CO_2}{O_2}$  Copyright © Save My Exams. All Rights Reserved <table border="1" data-bbox="858 271 1374 577"> <thead> <tr> <th>Respiratory substrate</th><th>RQ</th></tr> </thead> <tbody> <tr> <td>Carbohydrate</td><td>1.0</td></tr> <tr> <td>Lipid</td><td>0.7</td></tr> <tr> <td>Protein</td><td>0.9</td></tr> </tbody> </table> Copyright © Save My Exams. All Rights Reserved Respiratory Quotient: the ratio of carbon dioxide molecules produced to oxygen molecules taken in during respiration $RQ = \frac{\text{Moles/Volume of molecules of Carbon Dioxide given out}}{\text{Moles/Volume of molecules of Oxygen taken in}}$ Glucose: Carbon to oxygen ratio is 1:1 so RQ value is 1.	Respiratory substrate	RQ	Carbohydrate	1.0	Lipid	0.7	Protein	0.9
Respiratory substrate	RQ								
Carbohydrate	1.0								
Lipid	0.7								
Protein	0.9								
7.7	CORE PRACTICAL 15 Use an artificial hydrogen carrier (redox indicator) to investigate respiration in yeast.								
7.8	CORE PRACTICAL 16 Use a simple respirometer to determine the rate of respiration and RQ of a suitable material (such as germinating seeds or small invertebrates).								
7.9	know the way in which muscles, tendons, the skeleton and ligaments interact to enable movement, including antagonistic muscle pairs, extensors and flexors								
	Cartilage: hard but flexible skeletal tissue that often acts as a shock absorber and prevents wear in joints → Made of chondrocytes Ligament: Elastic tissue which joins bone to bone and forms joint capsule Tendon: Inelastic tissue which joins muscles to bones Synovial Joint: fluid which lubricates the most mobile joints  								

	Antagonistic Muscle Pairs - Muscles work antagonistically. - → This is because muscles can only pull, not push (sliding filament theory). - When pulling up the lower arm, it is flexed: biceps muscle (flexor) contracted and triceps muscle (extensor) relaxed - When lowering the lower arm, it is extended: biceps muscle (flexor) relaxed and triceps muscle (extensor) contracted
7.10	(i) know the structure of a mammalian skeletal muscle fibre (ii) understand the structural and physiological differences between fast and slow twitch muscle fibres
i	muscle muscle fibre transverse (T) tubule sarcolemma myofibril sarcomere I-band A-band M-line Z-line H-zone actin myosin cross-sections through the myofibril Muscles → Bundle of Muscle Fibres → Myofibrils → Sarcomere Sarcomere Structure: - Sarcoplasm: cytoplasm of myofibrils - Sarcoplasmic Reticulum: Stores and releases calcium ions - Actin - Myoglobin - A band: Contains H zone and area where myosin and actin filament overlap - H zone: Only thick myosin filaments present - I band: Only thin actin filaments present - M line: Attachment for myosin filament - Z line: Attachment for actin filaments and separates each sarcomere

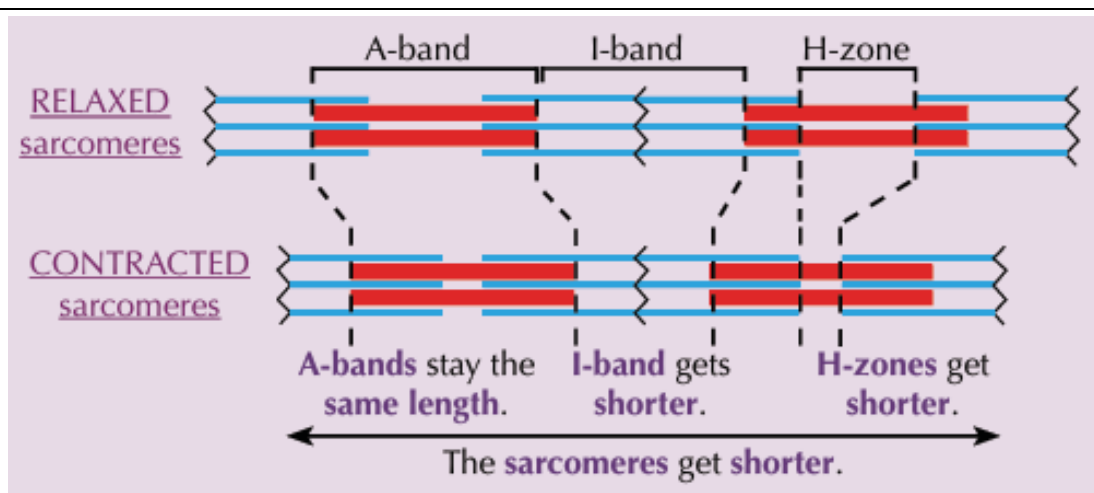
ii		
	Slow Twitch Muscles	Fast Twitch Muscles
	Energy supply relies on Oxidative Phosphorylation (Aerobic Respiration)	Energy supply relies on Glycolysis of Glycogen (Anaerobic Respiration) and Phosphocreatine
	1. Larger Capillary Network → Ensure good supply of oxygen and glucose for aerobic respiration	1. Fewer Capillary Network
	2. Many Mitochondria per Fibre → Allow more aerobic respiration	2. Fewer Mitochondria per Fibre Larger in diameter and more space so have more sarcoplasmic reticulum
	3. Many Myoglobin Molecules → Allows greater store of oxygen → Myosin molecule has red color so slow twitch muscle is more of red color	3. Fewer Myoglobin Molecules → So fast twitch muscles have paler color - Have more glycogen and creatine phosphate → For energy supply
	Slow and Sustained Muscle Contraction → Allows longer periods of exercise and decrease in fatigue → More cross-bridge formed in slow twitch contractions	Rapid and Intense → Allow to contract more quickly in a short time → Fatigue more quickly → Less cross-bridge formed in fast twitch contractions

7.11

understand the process of contraction of skeletal muscle in terms of the sliding filament theory, including the role of actin, myosin, troponin, tropomyosin, calcium ions (Ca^{2+}), ATP and ATPase

**Sliding Filament Theory of Muscle Contraction:**

1. Calcium ion channels open in sarcoplasmic reticulum
2. Calcium ions bind to troponin and cause myosin binding sites to be exposed
3. This allows myosin to bind to actin and actin-myosin cross bridge is formed
4. ADP and P_i are released, resulting in actin pulled over myosin towards the M line
5. ATP binds to myosin head and cross-bridge is broken as myosin head detaches
6. ATP is hydrolyzed to provide energy for myosin head to change shape and position
7. This allows myosin to bind to actin again



- 7.12**
- (i) know the myogenic nature of cardiac muscle
 - (ii) understand how the normal electrical activity of the heart coordinates the heartbeat, including the roles of the sinoatrial node (SAN), the atrioventricular node (AVN), the bundle of His and the Purkyne fibres
 - (iii) understand how the use of electrocardiograms (ECGs) can aid in the diagnosis of abnormal heart rhythms

i **Myogenic:** when muscle tissue that contracts and relaxes even when there is no stimulation from a nerve

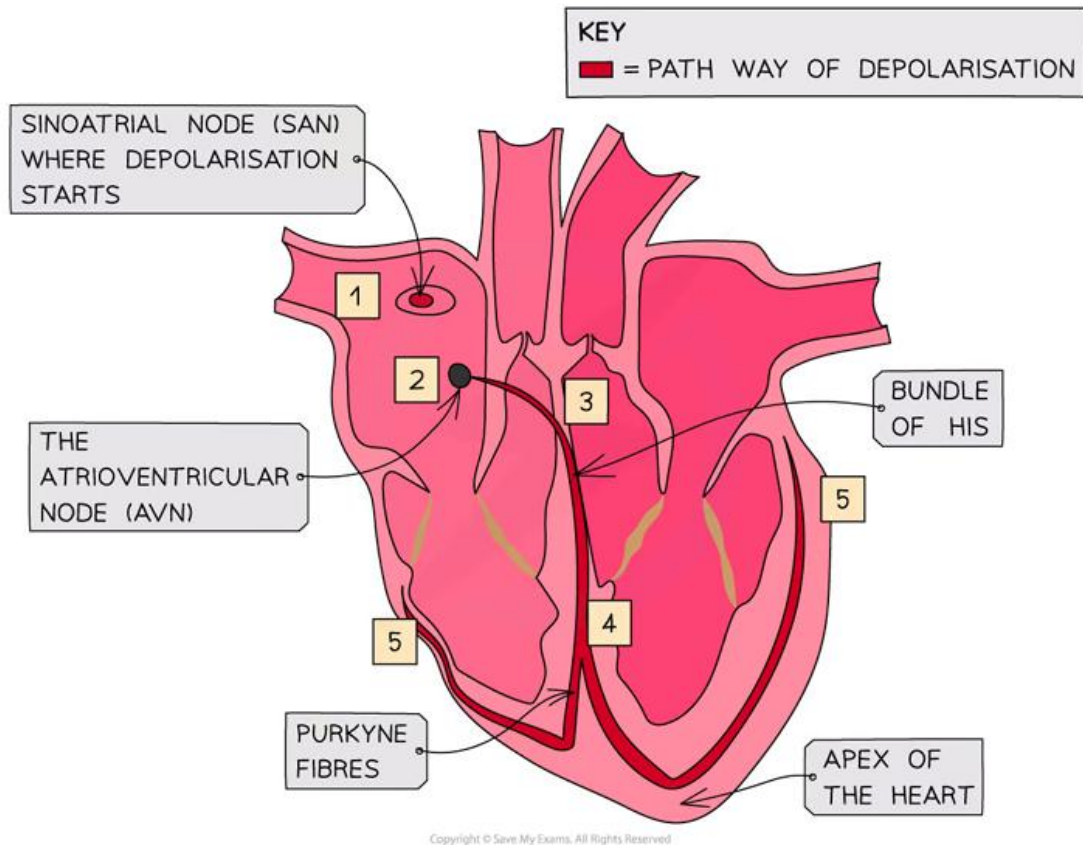
- **Cardiac muscles are myogenic.**

(3) Main Types of Muscles:

1. Skeletal Muscles
2. Cardiac Muscles
3. Smooth Muscles

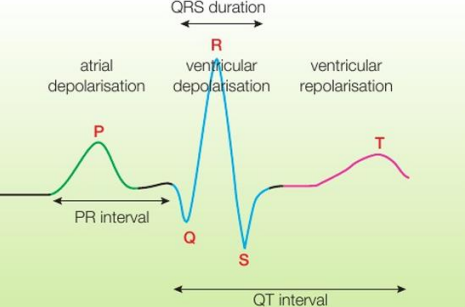
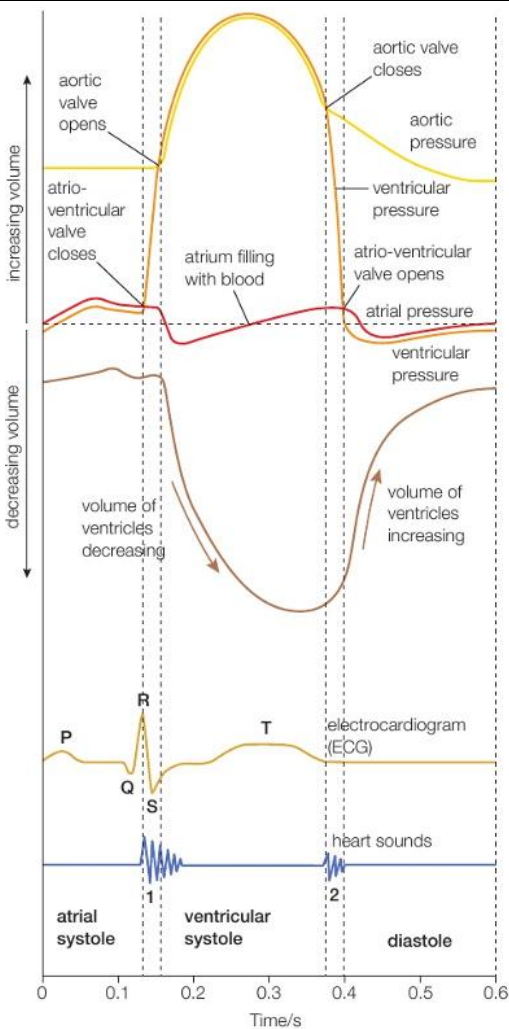
Features	Type of muscle		
	Striated	Cardiac	Smooth
Appearance in the light microscope	stripes (striations) at regular intervals	stripes (striations) at regular intervals	no striations
Cell structure	multinucleate (syncytium)	uninucleate cells joined by intercalated discs (Figure 8.22)	uninucleate cells
Shape of cells	long, unbranched cylinder	cells are shorter with branches that connect to adjacent cells	long, unbranched cells that taper at either end
Organisation of contractile proteins inside the cell	organised into parallel bundles of myofibrils	organised into parallel bundles of myofibrils	contractile proteins not organised into myofibrils
Distribution in the body	muscles attached to the skeleton	heart	tubular structures, e.g. blood vessels (arteries, arterioles and veins), airways, gut, Fallopian tubes (oviducts), uterus
Control	neurogenic	myogenic	neurogenic

ii



Maintaining Intrinsic Heart Rhythm:

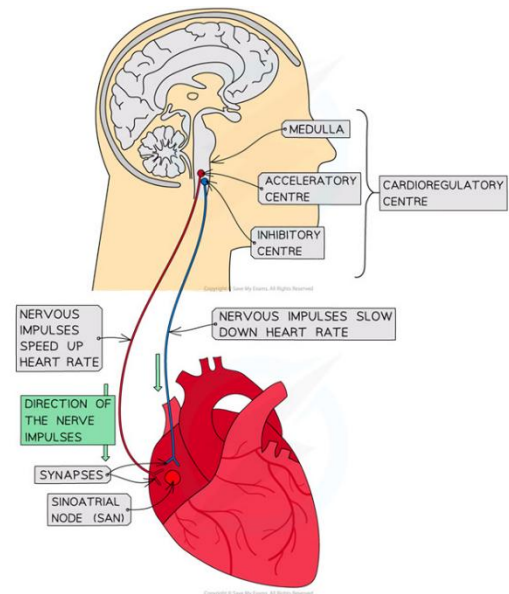
1. SAN (Sinoatrial Node) establishes a wave of electrical excitation (depolarization) which cause atria to start contracting
 → This initiates heartbeat
2. Excitation then spreads to AVN (Atrioventricular Node)
3. AVN is excited as a result of SAN but it produces a slight delay before the wave of depolarization passes into the bundle of His
 → This makes sure the atria have stopped contracting before the ventricle start
4. The bundle of His splits into two branches and carries the wave of excitation on into the Purkyne tissue
 → The Purkyne tissue consists of conducting fibres that penetrate down through the septum, spreading around the ventricles
5. As the depolarization travels through the tissues, it starts the contraction of the ventricles, starting at the bottom and so squeezing the blood out of the heart

iii	Electrocardiograph: is a machine that records the electrical changes in the heart using electrodes placed on the chest Electrocardiogram (ECG): the trace produced by an electrocardiograph P Wave: atrial depolarization; atrial systole QRS Region: ventricular depolarization; ventricular systole T Wave: ventricular repolarization; diastole When asked to compare (2) ECGs: - Compare P, R, T wave peak length - Compare duration between ST, PQ, and QRS 	 fig C Electrical and pressure changes in the heart during the cardiac cycle and the causes of the heart sounds.
7.13	(i) be able to calculate cardiac output (ii) understand how variations in ventilation and cardiac output enable rapid delivery of oxygen to tissues and the removal of carbon dioxide from them, including how the heart rate and ventilation rate are controlled and the roles of the cardiovascular control centre and the ventilation centre in the medulla oblongata	
i	$Cardiac\ Output\ (cm^3\ min^{-1}) = Heart\ Rate(beats\ per\ min[bpm]) \times Stroke\ Volume\ (cm^3)$ Why cardiac output provides more information than resting heart rate: - Cardiac output includes stroke volume - Cardiac output gives an indication of the efficiency of the heart → (It is best to compare cardiac output of different organisms rather than resting heart rate)	

ii

Controlling Heart Rate: (During Exercise)

- During exercise muscles use energy (ATP)
- Most of this energy is produced via aerobic respiration
- Heart transport bulk transport of oxygen and glucose
- Chemoreceptors detect lower pH in blood due to increased blood CO₂
- Baroreceptors detect change in blood pressure
- Electrical impulses sent to medulla oblongata
- Impulses travel from medulla oblongata via sympathetic nervous system to the SAN
- This causes SAN to increase frequency of depolarization causing an increase in heart rate
- Cardiac cycle is shorter

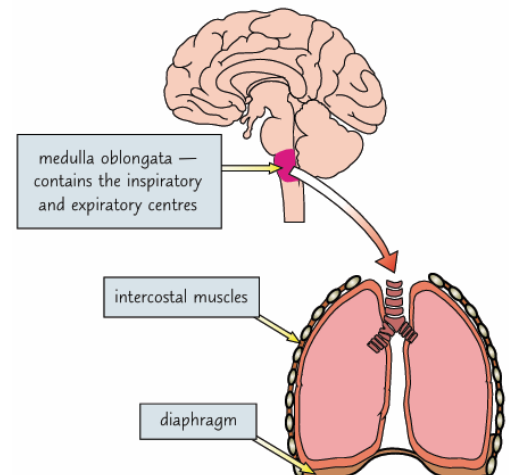


Following Exercise:

- As blood pH increases, more impulses are sent via the parasympathetic nerve
- Slowing the rate of impulses from SAN

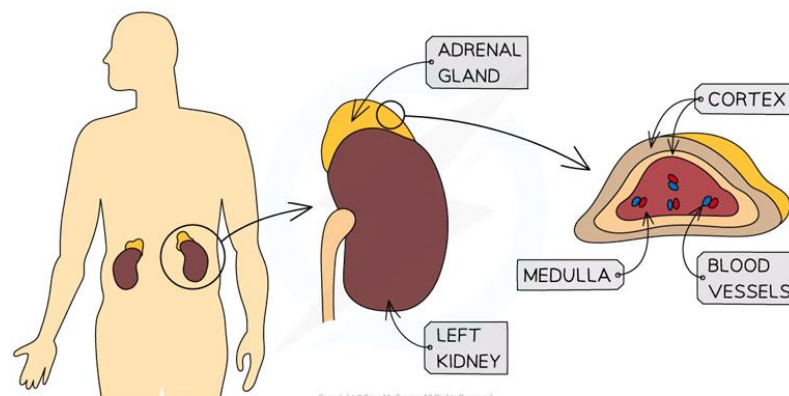
Controlling Ventilation Rate: (During Exercise)

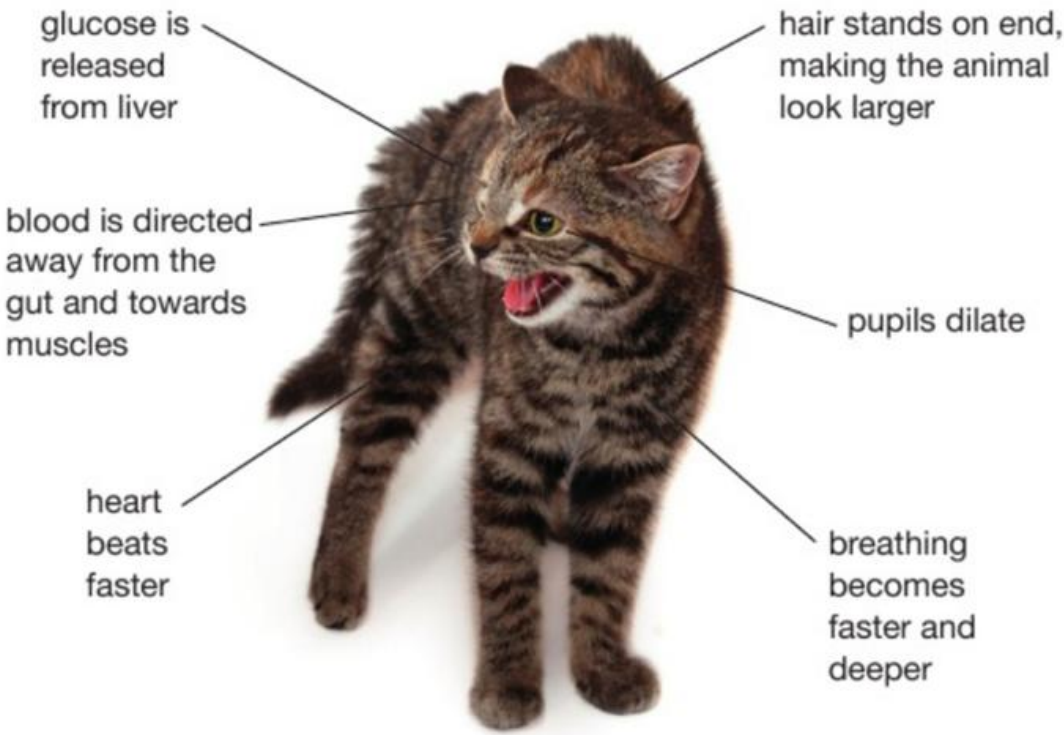
- Increase in activity leads to an increase in CO₂ which decreases pH
- This is detected by chemoreceptors
- This causes transmission of impulses to the ventilation center in medulla oblongata
- This then increases impulses via sympathetic neuron to increase contraction of intercostal muscles and diaphragm
- So, increase ventilation rate

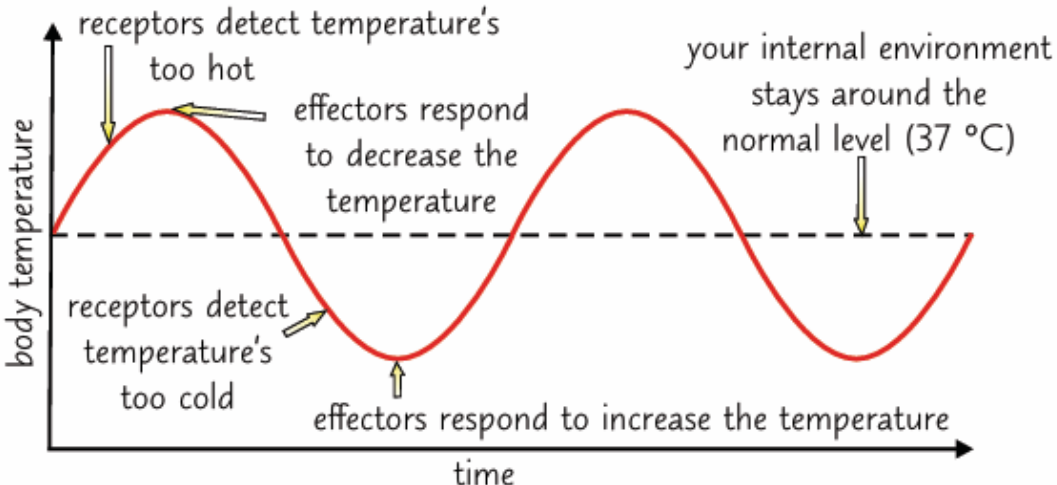
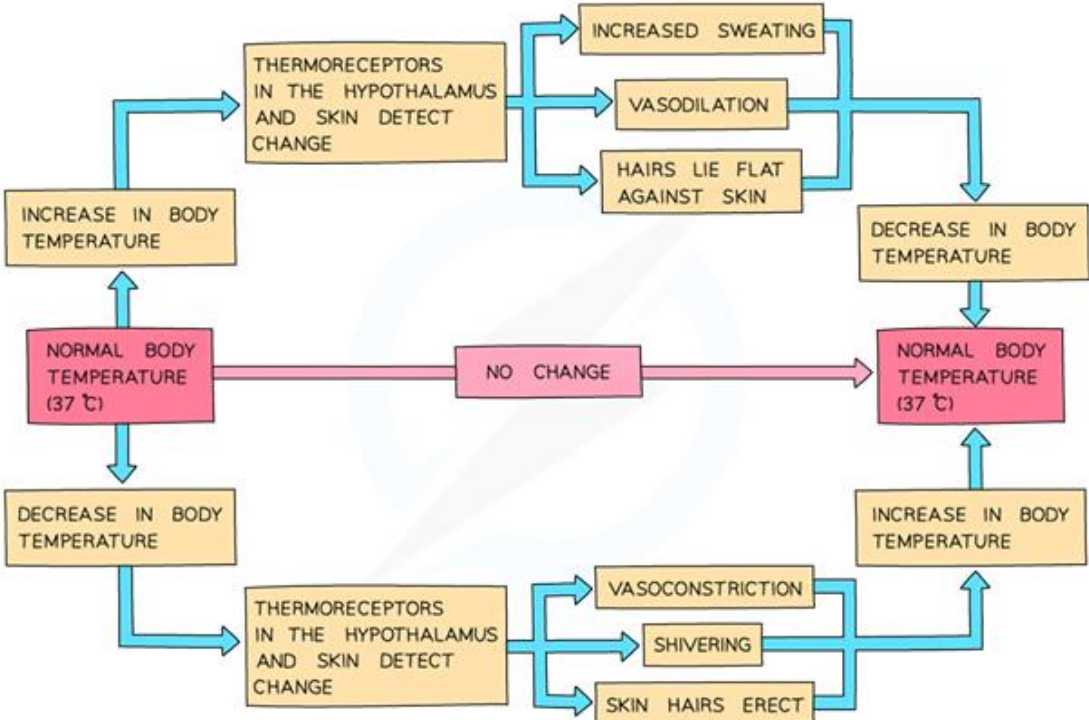


7.14

understand the role of adrenaline in the fight or flight response



	 ▲ Figure 7.4 Adrenaline affects the body of an animal in many ways. Adrenaline: a hormone that will prepare your body for 'fight or flight' response Effect of Adrenaline: - Cardiovascular Control Centre: adrenaline binds to SAN to depolarize more frequently which increases heart rate for faster rate of blood flow - Pupil Dilation: circular muscles relax and radial muscles contract to observe the environment more clearly - Digestive System: Blood vessels to digestive system constricts as it is diverted into skeletal muscles for faster contraction
7.15	CORE PRACTICAL 17 Investigate the effects of exercise on tidal volume, breathing rate, respiratory minute ventilation, and oxygen consumption using data from spirometer traces.
7.16	(i) understand what is meant by the terms <i>negative feedback</i> and <i>positive feedback control</i> (ii) understand the principle of negative feedback in maintaining systems within narrow limits
i	Negative Feedback: results in a decrease in the effect of the stimulus - Examples: Thermoregulation, Osmoregulation, Regulating Blood Glucose Level. Positive Feedback: results in an increase in the effect of stimulus - Examples: Dilation of Cervix during Labor, Blood Clotting, Depolarization of Membrane.

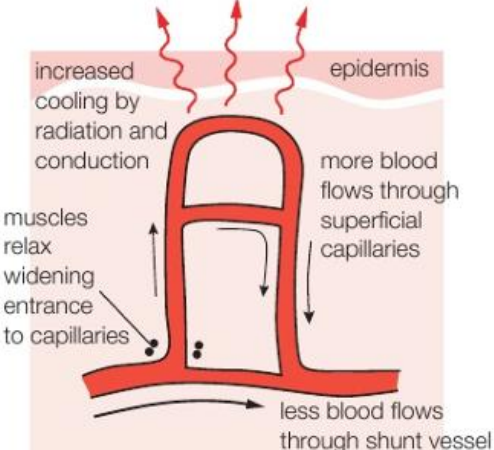
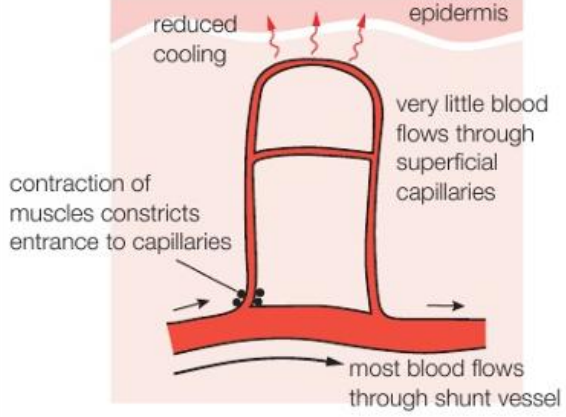
ii	Principle of Negative Feedback: e.g.: Stimuli (Change in blood temperature) → Receptor (or Sensor) → Neuron → CNS (Instruct corrective actions to reach set point) → Neuron → Effector (Produce Hormone) <u>Control of Body Temperature by Negative Feedback</u> 
7.17	understand what is meant by the term <i>homeostasis</i> and its importance in maintaining the body in a state of dynamic equilibrium during exercise, including the role of the hypothalamus in thermoregulation
	Homeostasis: Maintaining a state of dynamic equilibrium in the body, despite changes in the external or internal conditions  Copyright © Save My Exams. All Rights Reserved

Thermoregulation:

- Thermoreceptors detect changes in temperature
- Impulses sent to the thermoregulatory control centre in hypothalamus
- Hypothalamus sends impulses via sympathetic neurons to effectors
- Correct response by effectors to increase heat energy loss (during exercise): vasodilation, metabolize brown fat (thermogenesis), increased sweat production, erector pili muscles relax

Vasodilation Mechanism:

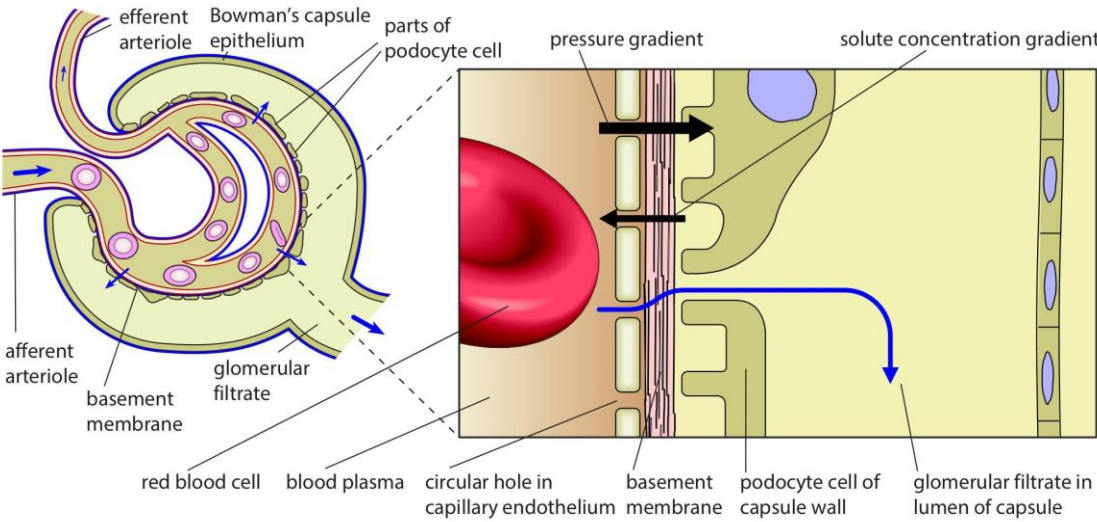
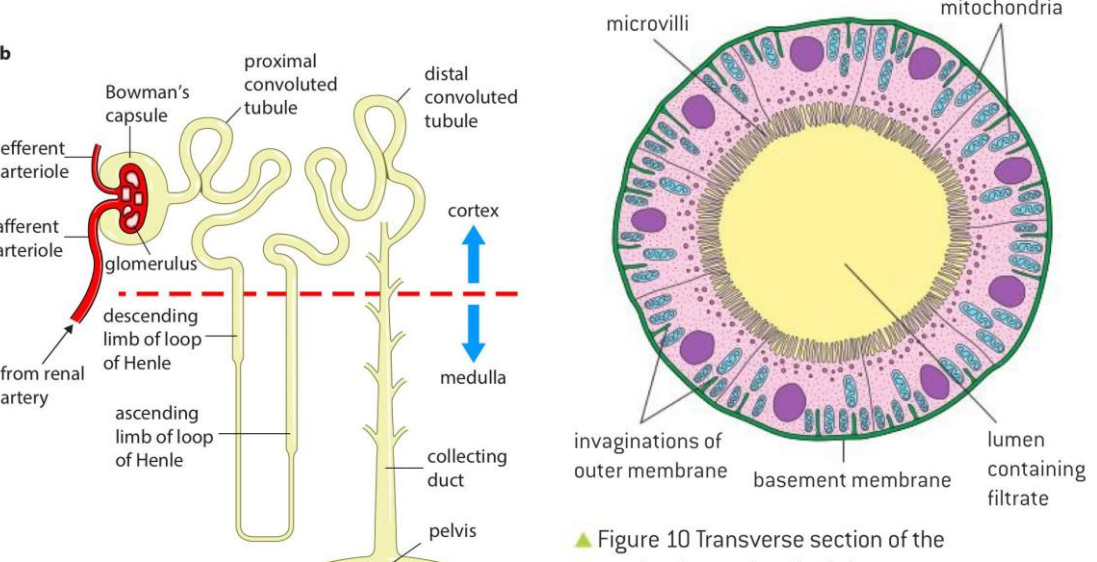
- Shunt vessels contract
- Sphincter muscles relax
- Blood flow to superficial capillaries

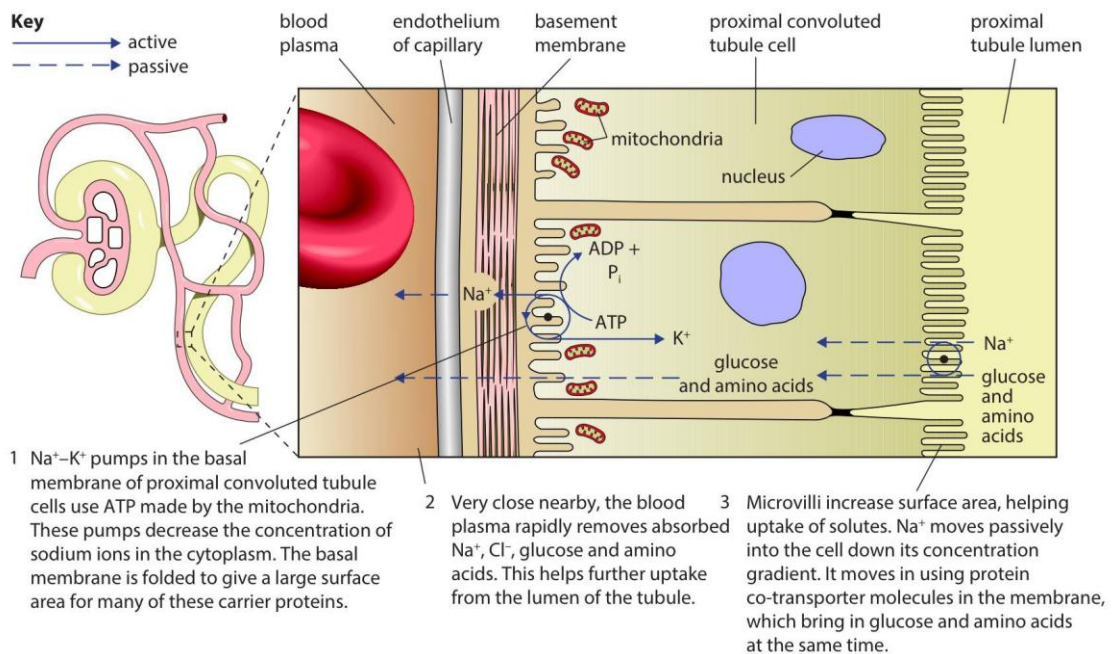
IN A WARM ENVIRONMENT	IN A COLD ENVIRONMENT
• vasodilation occurs 	• vasoconstriction occurs 
• sphincter muscles around the arterioles leading to the superficial capillaries are not stimulated to contract and therefore relax	• sphincter muscles around the arterioles leading to the superficial capillaries are stimulated to contract
• more blood can flow into these capillaries, dilating them (making them wider) with the pressure; less blood flows through deeper shunt vessels	• this constricts the passage of blood into these capillaries (making them narrower) so more blood flows through deeper shunt vessels
• more blood flows close to the body surface	• less blood flows close to the body surface
• as more blood flows close to the body surface, the temperature gradient between the body surface and the environment becomes greater, so cooling by conduction and radiation increases.	• as most blood is diverted further from the body surface, the temperature gradient between the body surface and the environment is lower, so cooling by conduction and radiation decreases.

7.18

know the gross and microscopic structure of the mammalian kidney

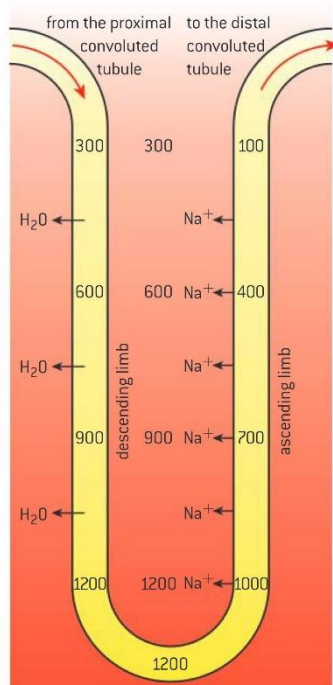
	fig C (a), (b) and (c) The gross structure of the mammalian kidney as seen with the naked eye. (d) The basic structure of the mammalian kidney. The position of the two main types of kidney tubule (see below) have been superimposed on this diagram. Kidney has (2) main roles: 1. Excretion 2. Osmoregulation → Osmoregulation has (3) main functions: 1. Ultrafiltration 2. Selective Reabsorption 3. Tubular Secretion
7.19	understand how urea is produced in the liver from excess amino acids (<i>details of the ornithine cycle are not required</i>) and how it is removed from the bloodstream by ultrafiltration
	Deamination Process to produce Urea - Body cannot store excess protein or amino acids - Liver cells, hepatocytes, are responsible to remove the amino group from excess amino acids in a process called deamination - Keto acid enters the Krebs cycle for aerobic respiration or converted to glycogen - Ammonia produced is toxic so it is combined with carbon dioxide into urea and water in ornithine cycle - Urea is filtered out of the bloodstream into Bowman's capsule of the nephron by ultrafiltration $\begin{array}{c} \text{R} \\ \\ 2\text{NH}_2 - \text{C} - \text{COOH} + \text{O}_2 \rightarrow \begin{array}{c} \text{R} \\ \\ 2\text{C} = \text{O} - \text{COOH} \end{array} + 2\text{NH}_3 \\ \\ \text{H} \\ \text{amino acid} \qquad \text{oxygen} \qquad \text{keto acid (to Krebs cycle)} \qquad \text{ammonia} \end{array}$ ammonia feeds into ornithine cycle

	 How Urea is removed by Ultrafiltration: - Urea is forced out by high pressure of the blood - This is caused by afferent blood vessels greater diameter than efferent blood vessel in the glomerulus - So, urea flows through pores in the basement membrane
7.20	understand how solutes are selectively reabsorbed in the proximal tubule and how the loop of Henle acts as a countercurrent multiplier to increase the reabsorption of water
	 ▲ Figure 10 Transverse section of the proximal convoluted tubule



Selective Reabsorption in Proximal Tubule:

- Glucose reabsorption takes place in proximal tubule
- This is active transport in the membrane of nephron through channel proteins and carrier proteins
- Also, co-transport with sodium ions
- Diffuse into blood capillaries down the concentration gradient



▲ Figure 12 Solute concentrations in the loop of Henlé (in mOsm)

How Loop of Henle acts as a Countercurrent Multiplier:

- Countercurrent: Flows of fluid in opposite direction
- Multiplier: It causes a steeper gradient of solute concentration to develop in the medulla

1. Filtrate in the two limbs of the loop of Henle moves in opposite directions
2. Sodium and potassium ions move out of the ascending limb
3. Ions move by active transport
4. Resulting in decrease in ion concentration of surrounding fluid (in medulla)
5. Therefore, water leaves the descending limb by osmosis

- **Descending limb** is permeable to water but not very permeable to sodium and chloride ions
- **Ascending limb** is very permeable to sodium and chloride ions but not permeable to water

Overall Process happening in a nephron:

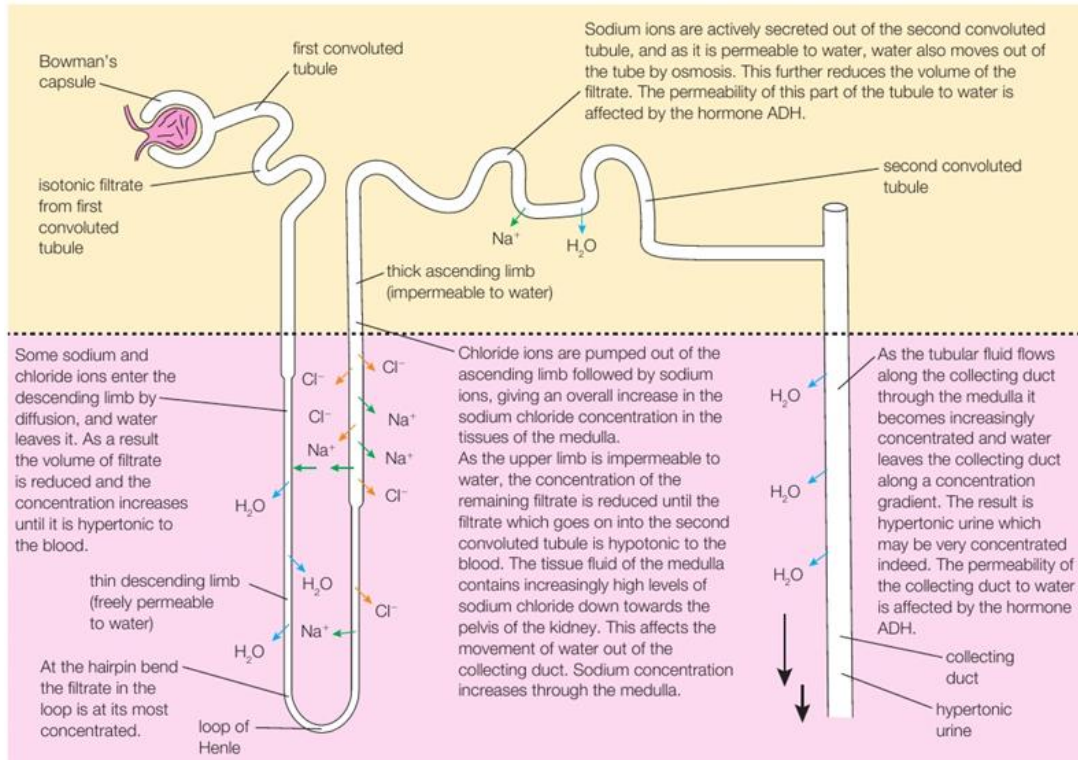
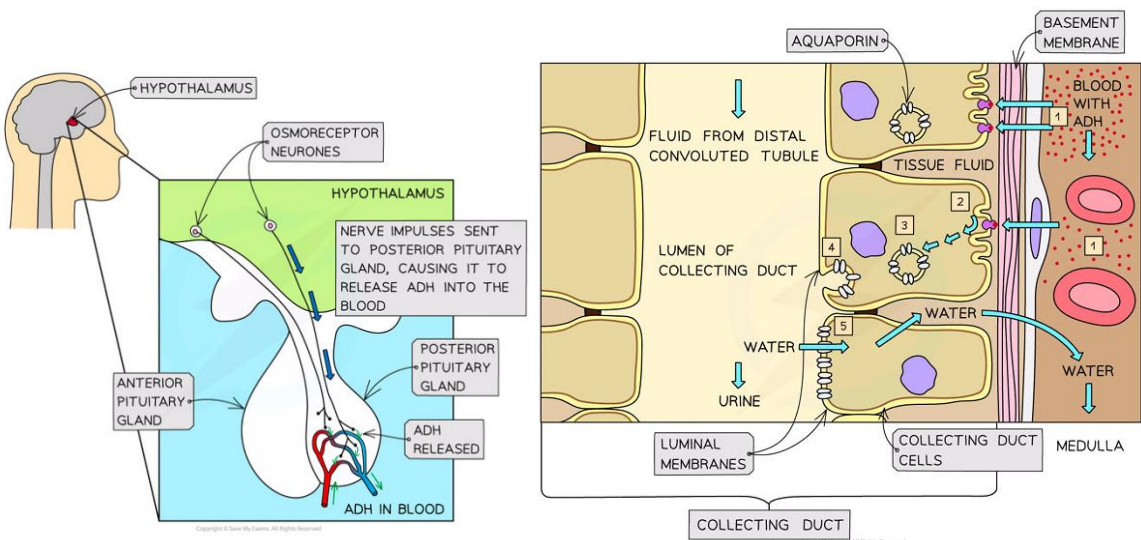


fig E A model of the role of the loop of Henle in the reabsorption of water and the production of concentrated urine in the kidney

7.21 understand how the pituitary gland and osmoreceptors in the hypothalamus, combined with the action of antidiuretic hormone (ADH), bring about negative feedback control of mammalian plasma concentration and blood volume



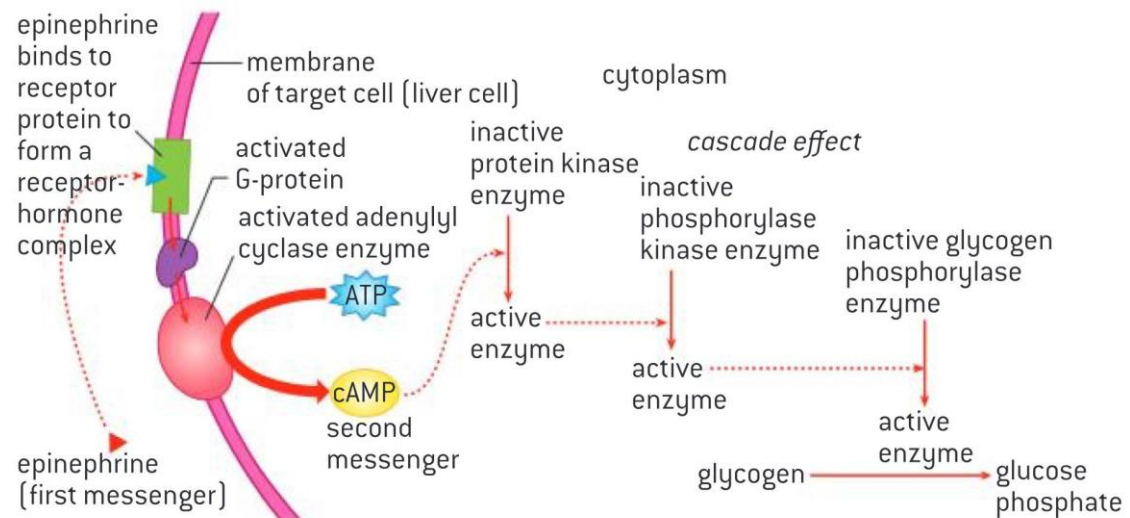
How ADH is involved in control of water potential of the blood:

- Water potential of blood is monitored by osmoreceptors in hypothalamus
- Hypothalamus produces ADH and stores into pituitary gland
- Pituitary gland releases ADH
- In blood to kidney, ADH travels to act on collecting duct and distal convoluted tubule
- This inserts aquaporins into cell membranes of collecting duct
- This increases in permeability of water of cell membranes
- More water reabsorbed so water potential of blood is back to normal range

7.22

understand how genes can be switched on and off by DNA transcription factors, including the role of peptide hormones acting extracellularly and steroid hormones acting intracellularly

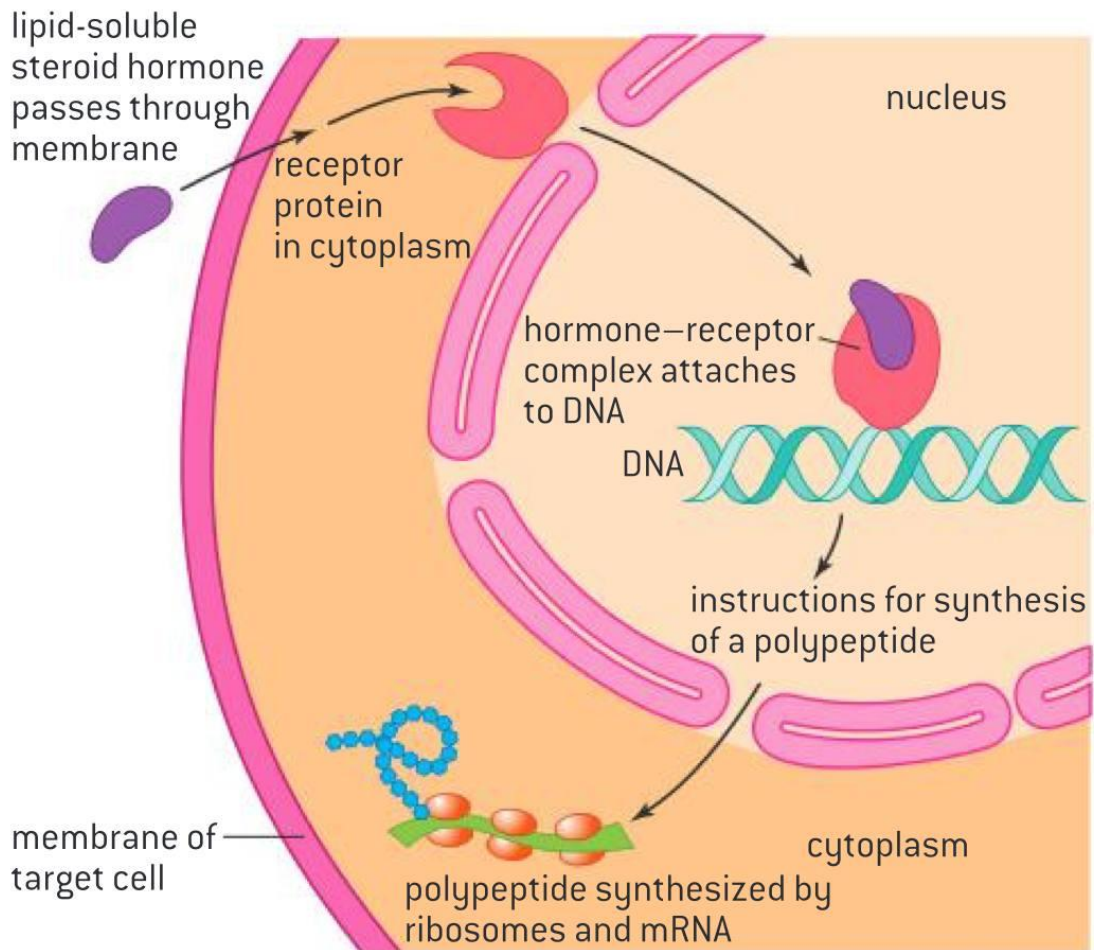
(2) Types of Hormones:



▲ Figure 5 Mechanism of action of adrenaline on a liver cell

1. Peptide Hormones:

- Peptide hormones are not soluble in lipids so cannot cross the cell membrane
- They bind to receptors in the plasma membrane of target cell
- This activates a cascade mediated by second messenger inside the cell
 - Second messengers are small water molecules that can quickly spread throughout the cytoplasm and relay signals throughout the cells
 - The second messenger activates protein kinase enzymes which trigger a chain of reactions called cascade



▲ Figure 4 Mechanism of action of steroid hormones

2. Steroid Hormones:

- Steroid hormones can cross directly through the plasma membrane and nuclear membrane as they are lipid soluble
- They bind to receptors to form receptor-hormone complex
- This acts as DNA transcription factor, regulating gene expressions by switching sections of DNA on or off

Transcription Factors

- Transcription factors are proteins
- Genes need to be switched on before proteins can be synthesized
- Transcription factors bind to specific regions of DNA / promoter sequences
- This catalyzes methylation / demethylation of DNA or acetylation / deacetylation / dephosphorylation of histone proteins
- So, RNA polymerase binding can be controlled
- **Switching on** gene:
 - Open chromatin
 - Demethylation of DNA
 - Acetylation of histone (leads to change in ionic charges in histone)
- **Switching off** gene:
 - Condensed chromatin
 - Methylated DNA
 - Deacetylated histone

Notice

- In this booklet, mark scheme points from past paper questions (mostly 2020-2023 and a few from 2010-2019) are included for each specification point to connect the syllabus and the past paper questions.
- For less frequently examined specification points, Pearson Edexcel IAL textbook and Save My Exams (available at [savemyexams.com](https://www.savemyexams.com)) notes have been the main reference of this booklet.
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